



Hyperpurex®

FX

Flagship series

Intelligent medium lab pure water system



Breakthrough and Innovation All New Masterpiece



Hyperpurex[®]

FX Flagship series (FXU/FXUS/FXUE/FXDE)

Intelligent medium lab pure water system

Integrated all-plastic box body, featuring FX-pure water production system with 200-liter PE pure water tank and PWS-intelligent storage and distribution system, to highlight the aesthetics of science and technology through breakthrough design.

FX-pure water production system, using the innovative intelligent human-computer interactive control system and 7-inch colorful capacitive touch screen, integrating functions of Internet of Things (IOT) and cloud platform, embedding new purification cartridges with patented structure, rigorous double RO system^[1], advanced EDI module^[2] and DI ion-exchange cartridges with larger capacity.

System output: 50-250 liters/h, It can simultaneously produce ultrapure water (18.2MΩ.cm), high pure water (>17.5MΩ.cm) and pure water (<5μs/cm)^[3]. The quality of pure water fully meets or exceeds the requirements of water quality standard specified by ASTM D1193-06, GB/T 11446.1-2013, GB/T 33087-2016, GB/T 6682-2008, CP, EP, USP, JP, CAP, CLSI, etc. Optional PWS-intelligent storage and distribution system module, compatible with various accessories, delivers pure water through lab pure water pipelines at specific flow rates and pressures, meeting the overall pure water needs of multiple lab or entire lab building.

^[1] The RO^{2nd} system is only used for FXUS/FXUE/FXDE series products.

^[2] EDI module is only used for FXUE/FXDE series products.

^[3] RO water (<5μs/cm) is only applicable to RO^{2nd} system.

Application Area:

- HPLC、UPLC、LC-MS
- ICP-MS、ICP-AES、AAS、GC-MS
- MALDI-TOF-MS、IC、TOC analysis
- Electrochemical, spectrophotometric determination
- Preparation of microbial media and reagents
- Cell culture, PCR, IVF
- Protein purification, electrophoresis, biochemistry
- Proteomics, genomics, immunoassay
- Feed water of laboratory instruments, such as: autoclave, bottle washing machine, environmental test chamber, water bath, etc.

Innovative Control System Bringing Efficiency & Intelligence

All new intelligent human-computer interactive control system

- 7 inch colorful capacitive touch screen, with glass panel, achieve touch operation experience, the same as mobile phone.
- LINUX system, resolution:1024×600, free switching in dual language (Chinese-English) display and 3 kinds of theme interface.
- Simple and intuitive UI interface design, to display the newly defined cartridge, storage and water quality status. So that you can fully understand the system operating status and parameters. A glance for all important information.

Comprehensive water quality monitoring and alarm

- 4 water quality sensors, to monitor water quality and alarm (Feed water, RO water, DI water and UP water), electrode constant-0.01cm⁻¹, temperature sensitivity-0.1 °C , and the conductivity/resistivity after temperature compensation and water temperature can be displayed simultaneously.
- 4 flow sensors, including 3 channels (RO water, DI water and UP water) quantitative dispensing and 1 channel of feed water accumulation function.
- With real-time display and alarm function of ion rejection rate of RO membrane.
- Equipped with feed water pressure sensor, real-timely monitor and display inlet pressure, convenient to timely understand the feed water condition.
- Built-in TOC detection module, real-timely monitor TOC of ultrapure water, detection range: 0.5-999.9ppb, detection accuracy:±0.1ppb, in line with USP and EP system adaptability test^[1].

^[1] Applicable to ultrapure water systems equipped with UV module.

Sharing mode of multi-user management

- The main user can add multiple secondary users. Each user has an independent ID and password.
- The lock screen key can achieve the rapid locking and user switching of the device, allowing multiple departments and projects to share the device.
- Function of water dispensing record report, easy to view and export the water dispensing record of each ID, achieve the allocation of cartridges and maintenance costs according to the user ID.

Innovative Internet of Things (IOT) and cloud platform technology

- Access the internet by Ethernet and WIFI, achieve remote data acquisition, monitoring and management.
- Log into the cloud platform from PC, WAP or WeChat to get the device information.
- Timely alarm information to achieve fast customer service response.
- Health analysis based on big data makes fault judgment more accurate.
- Work order system can provide customers with efficient service guarantee.
- It can be connected to LIMS or BMS to realize equipment informatization and make laboratory information management more efficient and standardized.

Traceable comprehensive data management

- Store operating data records, up to 5 years, including water dispensing, alarm and cartridge replacement, achieve data storage and paperless data management of the whole product life cycle by the cloud platform, and meet data tracking needs.
- With function of data exporting from USB port and data downloading from cloud platform.
- Data report documents in PDF format can prevent tampering, meeting traceability provisions of data integrity.
- Function of water dispensing report, including each water quality, water volume and user information, meet the regulatory requirements, easier to certificate.

Flexible and diverse water dispensing mode^[1]

- One system with 4 water dispensing points, 3 kinds of pure water - RO water, DI water and UP water, to meet all lab water needs.
- The host is equipped with 3 (RO/DI/UP) standard pure water outlets, general and quantitative - 2 kinds of water dispensing mode, bringing all new dispensing experience.
- Professional-grade water tank provides the 4th pure water outlet, greatly improving water dispensing efficiency.

^[1] According to different model, pure water kind and dispensing mode are different. For details, refer to the product manual.



Perfect consumables management

- Function of consumable life management (PP/PC/RO/DI/EDI/UP/UV/UF/TF[1]) combined with water quality, time and capacity, to reduce consumable costs.
- The remaining life of cartridges will display on the main interface with the form of dynamic icon percentage. Function of cartridge life reminder, step by step with 3 kinds of color, blue (normal), yellow (warning), red (alarm), all cartridges status is clearly visible.
- Click the cartridge icon to view the cartridge status, purification technology and cartridge function, and can realize the function of scanning code to buy the cartridge.
- Serial number verification function of original cartridges, encrypted long serial number verification code, to prevent misoperation of cartridges installation and replacement.

^[1] According to different model, cartridges are different. For details, refer to the product manual.

Powerful system overview diagram

- Dynamically display process flow, cartridges configuration, water quality parameters, RO rejection rate, cartridges life and tank storage status, etc.
- Button of RO flushing, UF flushing[1] and system disinfection function[2] can start the forced cleaning of RO membrane, UF membrane and the system disinfection with chemical dosing, and remaining time of 3 procedures above is real-time display. The flushing function significantly prolongs the service life of the RO membrane and UF membrane, and the disinfection function effectively sterilizes the pure water pipeline system.
- Ultrapure water circulation system with adjustable interval running times, to keep the system in low levels of bacterial contamination and reduce energy consumption.

^[1] Applicable to ultrapure water systems equipped with UF module.

^[2] Applicable to ultrapure water system.

Intuitive tank storage status

- Use advanced pressure sensor to measure liquid level. The dynamic icon on the main interface can display the tank liquid level, storage capacity (L) and storage percentage (%) in real time. A glance for storage status.

Full security protection

- 3 level permission management, administrator and ordinary users have strict permission distinction.
- Integrated double water leakage protection and alarm device, to monitor water leakage inside the machine and on the ground.
- With alarm protection of No feed water, low feed water pressure, system high pressure and full tank.
- With alarm function of standard-exceeding of Feed/RO/DI/UP water quality, and end of cartridge life.
- All alarm information can be stored in the host and cloud platform, to meet data security requirements.

Advanced System Configuration Guaranteeing Strong Quality



1 Powerful 20-inch pretreated cartridge

- PP cartridge with deep folding membrane, accuracy of 5μm, to filter particles efficiently in source water.
- PC cartridge with high performance activated carbon fiber with catalyst, accuracy of 5μm, to adsorb organics and residual chlorine efficiently and avoid carbon powder precipitation maximumly.
- The combination of folding filter & carbon fiber filter can bring greater cartridge processing capacity, extend the replacement cycle and reduce the running cost.



3 High performance purification cartridge

- Patented cartridge structure can prevent the stratification of resin and ensure the exchange capacity of cartridge.
- The resin filling capacity per cartridge is up to 5.2 liters, and up to 2 cartridges^[1] can be equipped every host, with a total filling capacity of 10.4 liters, achieving greater ion exchange capacity and significantly reducing the running cost.
- All DuPont resin and high purity material of column ensure absolute 18.2MΩ.cm of ultrapure water resistivity and reduce TOC precipitation.

^[1] When system output ≥ 150 l/h, the purification resin cartridge uses external super large DI cartridge with capacity of up to 32L resin filling.



2 Rigorous double RO system^[1]

- Double RO system can remove up to 99% soluble inorganic ions, 99% soluble organics, microorganisms and particles.
- Compared with single RO system, the double RO water quality can be stable < 5μs/cm (feed water conductivity < 1500μs/cm), and the life of the ultrapure unit is longer.
- Equipped with high quality industrial grade RO membrane, to achieve combination of long life, stability and high ion rejection rate.
- Auto-flushing function of RO module with adjustable flushing interval and duration, to effectively prevent scale and prolong the life of the membrane.
- The automatic discharge function of unqualified RO water can ensure that the RO water quality is suitable to enter the back-end module.
- Integral package of discarded RO module, easy to install and maintain.

^[1] Applicable to FEUS/FEDS/FERS.



4 Professional mini EDI module^[1]

- Without softener and chemical regeneration, pure water in grade II, with resistivity > 10MΩ.cm@25 °C (generally above 15MΩ.cm), and TOC<30ppb^[2], is available. It can prolong the life of back-end module and reduce the running cost, reducing the running cost.

^[1] Applicable to FXUE/FXDE.

^[2] The values vary depending on the nature and concentration of contaminants in source water.



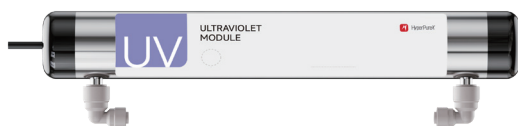
5 Built-in 7.5-liter pressure water tank

- With dual functions of water storage and pressurization, FDA approved, its fully enclosed structure effectively isolates air, and prevent the touching of CO₂ and other pollutants with pure water. Up to 100 liters is optional volume.
- Optional 250 liters professional PE pure water tank, equipped with liquid level sensor and air filter, to meet the needs of large water consumption.



6 250 liters professional PE pure water tank

- Integrated design with FX host, both pure water tank and main box, compact and exquisite structure
- HDPE material, double layer design, the outer layer with anti-UV inhibitor, to prevent the growth of algae inside and improve the durability of the tank, the inner layer with pure PE raw material, to reduce material precipitation and ensure water quality safety.
- Drainage valve, installed at the cone bottom, which can empty the water tank and ensure thorough cleaning.
- Feeding from the bottom can reduce CO₂ absorption.
- The enlarged cover with seal, to prevent air from entering and facilitate manual cleaning.
- Liquid level pressure sensor, in the standard configuration, with an accuracy of 1%, continuously displays liquid level in the water tank.
- Compound air filter, in the standard configuration, containing special packing and microporous membrane, to absorb CO₂ and organics, and filtrate bacteria and particles.
- Optional UV disinfection module, to sterilize tank and inhibit the growth of bacteria in the tank.



7 Double wavelength UV module^[1]

- Long-life ultraviolet lamp (185&254nm), combined with SUS316L flow shell, can reduce the value of TOC to $\leq 2\text{ppb}^{[2]}$, and can achieve efficient sterilization and inhibit bacterial growth, suitable for HPLC, UPLC, LC-MS and other precision instruments.

^[1] Applicable to ultrapure water systems equipped with UV module.

^[2] The values vary depending on the nature and concentration of contaminants in source water.



8 Ultrafiltration module^[1]

- With PES membrane and MWCO>5000D, effectively removes pyrogen/endotoxin, RNase, DNase, and produces nuclease-free, proteinase-free and bacterial-free ultrapure water, suitable for life science applications, such as cell culture/IVF.

^[1] Applicable to ultrapure water systems equipped with UF module.



9 MF terminal microfilter^[1]

- (0.45±0.2)μm double-layer PES membrane ensures microbial retention, effectively removes particles and bacteria, and meets critical application requirements.

^[1] Applicable to ultrapure or high-pure water system. For details, refer to the product manual.



11 Multiple communication interfaces^[1]

- USB interface, to export running data or upgrade system version online.
- RJ45, USB/WIFI interface, to achieve the IOT and cloud platform connection, easy access to LIMS or BMS.
- HiDis pure water dispensing arm interface, to achieve power supply and data communication with the host.
- L-Tank pure water tank interface, to synchronize water tank level signal with the host.
- FS foot switch interface, easy to dispense, suitable for more dispensing scenarios.
- LS leakage sensor interface, timely alarm external water leakage, strengthen equipment safety.

^[1] The interface configuration varies according to the model. For details, refer to the product manual.



10 UF terminal ultrafilter^[1]

- With PES membrane and MWCO>15000D, effectively removes pyrogen/endotoxin, RNase, DNase, and produces nuclease-free, proteinase-free and bacterial-free ultrapure water, suitable for life science applications, such as cell culture/IVF.

^[1] Optional accessory for ultrapure water system only.

FX Combined with PWS

Achieve the Whole Pure Water Workstation

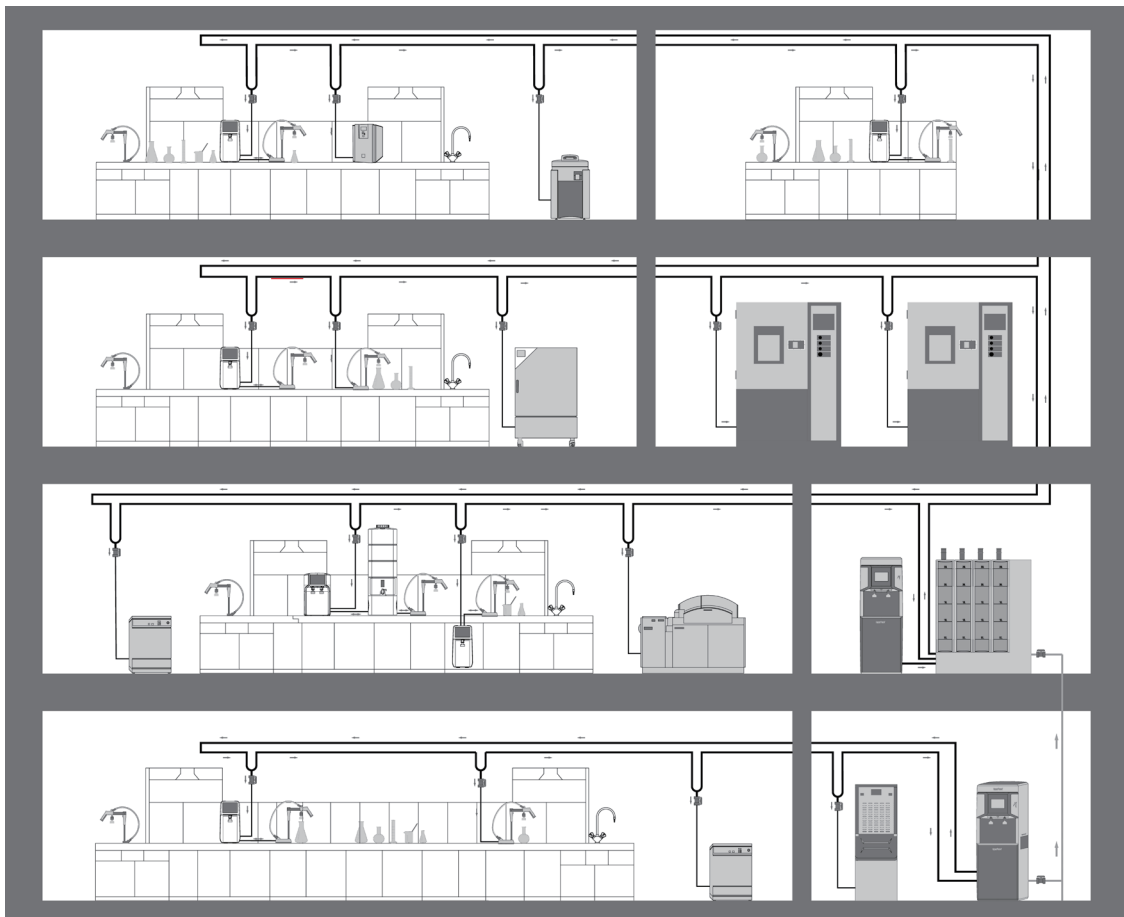


Optimal selection of medium and small lab central pure water system

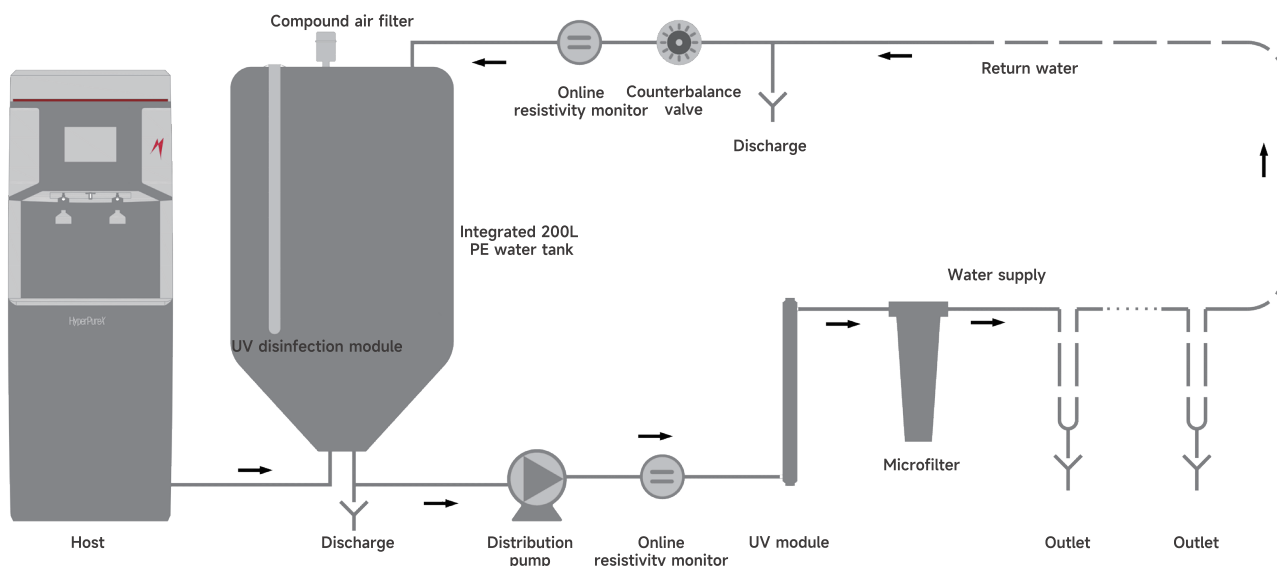
- Small and medium-sized lab central pure water supplying system, designed for single, multiple laboratories or single-storey laboratory building.
- FX water production host (0.4m²) and PWS-L distribution system (0.3m²), less than 0.7m² of total floor area, no need to reserve a dedicated pure water room, save valuable laboratory space.
- The noise reduction process and silent design of the all-plastic box, can be directly put into the lab to avoid disturbing your focusing on the experimental task.

Highly integrated all-in-one design

- Integrating all purification modules from the FX-pure water production system to the PWS-intelligent storage and distribution system (including raw water pretreatment, RO host, EDI and ultra-purification modules, as well as 200-liter tank, disinfection components, distribution pumps and distribution modules, etc.), one system meets all your pure water application needs.



Central Pure Water System Floor Plan



Intelligent PWS-L storage and distribution system



- PLC intelligent control, to achieve fully automatic distribution circulation and dosing disinfection functions, and can continuously optimize control programs according to different pure water pipeline layouts and water usage situations, to ensure high-quality pure water of the system.
- Distribution cycle function, which can set multiple working modes such as day/night, workday/holiday, etc., to achieve the combination of energy saving and ensuring high-quality water quality of pipelines.
- Dosing disinfection function, which can use Hyperpurex® exclusive disinfectant, to control low levels of bacterial contamination throughout the system.

Pure water-storage module

- Integrated 200L PE water tank, integrated design with the main host.
- Liquid level pressure sensor, in the standard configuration, with an accuracy of 1%, continuously displays liquid level in the water tank.
- Compound air filter, in the standard configuration, containing special packing and microporous membrane, to absorb CO₂ and organics, and filtrate bacteria and particles.
- Optional UV disinfection module, to sterilize tank and inhibit the growth of bacteria in the tank.

Pure water-distribution module

- Includes: distribution pump, UV lamp module in the pipeline, 0.2μm microfilter and resistivity monitor in outlet and return water pipeline, pressure regulating valve and high-quality pipeline, valves and adapters, to avoid microbial contamination and ensure long-term stable operation of the system.
- Optional water dispenser in the user points, which can monitor the quality of pipe outlet in real time.
- Optional UF ultrafiltration module in the user points, which can effectively remove pyrogen/endotoxin, RNase, DNase, to meet the requirements of life science applications such as cell culture /IVF.



Continuously upgraded PWS-M storage and distribution system

- Equipped with N+1 250-liter PE water tank, to achieve central water supplying of big water consumption in multi-storey or the whole experimental building, and according to the needs of experimental development, it can continuously increase the water tank and upgrade the system to meet new pure water needs.
- PWS-M has the same PLC intelligent control, automatic distribution circulation and dosing disinfection as PWS-L, which is the advanced choice when the 200-liter water tank of PWS-L system cannot meet your application needs.

FXU Flagship series

Intelligent Medium Ultrapure Water System

—Ultrapure water, high pure water, and RO^{1st} water

Integrated all-plastic box body, with a built-in professional-grade 200-liter PE pure water tank.

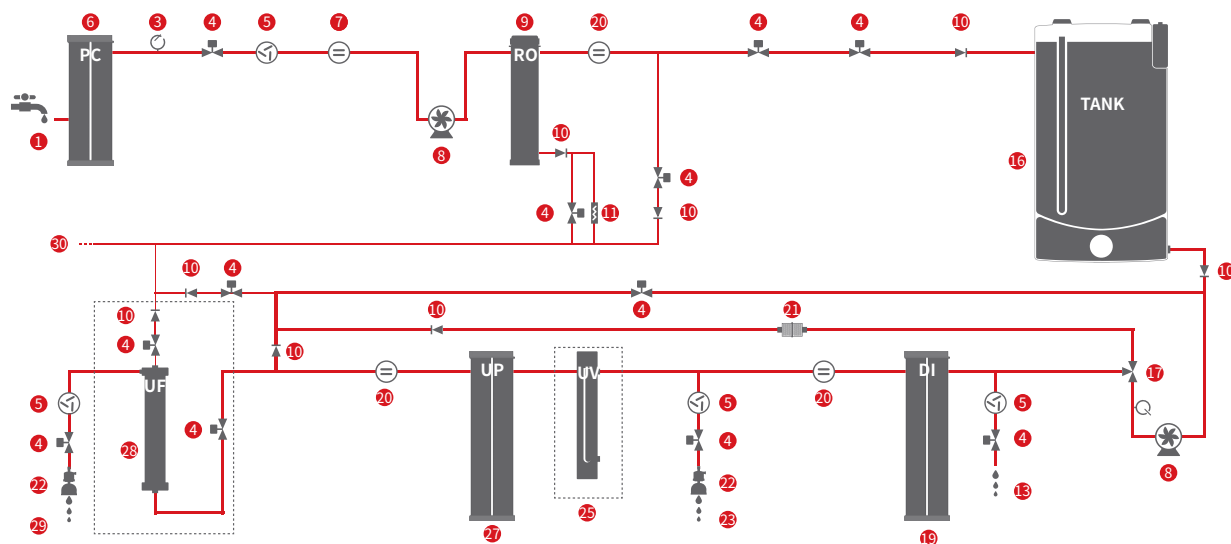
With tap water inlet, using the innovative intelligent human-computer interactive control system and 7-inch colorful capacitive touch screen, integrating functions of Internet of Things (IOT) and cloud platform, embedding new purification cartridges with patented structure, stable and reliable single RO system, and DI ion-exchange cartridges with larger capacity.

System output: 60, 120, 180, 250 liters/h. It can simultaneously produce ultrapure water (18.2MΩ.cm), high pure water (>17.5MΩ.cm) and single RO water. The quality of pure water fully meets or exceeds the requirements of water quality standard specified by ASTM D1193-06, GB/T 11446.1-2013, GB/T 33087-2016, GB/T 6682-2008, CP, EP, USP, JP, CAP, CLSI, etc.

Optional PWS-intelligent storage and distribution system module, compatible with various accessories, delivers pure water through lab pure water pipelines at specific flow rates and pressures, meeting the overall pure water needs of multiple lab or entire lab building.



Flow Diagram



- | | | | |
|-----------------------------|-----------------------|-----------------------|---------------------------|
| ① Feed Water | ⑨ RO cartridge | ⑰ Three way valve | ②⑤ UV Component |
| ② PP Pretreatment Cartridge | ⑩ One way valve | ⑱ High tension switch | ②⑥ TOC Component |
| ③ Pressure sensor | ⑪ Flow Restrictor | ⑲ DI Cartridge | ②⑦ UP Ultrapure cartridge |
| ④ Solenoid valve | ⑫ Pressure water tank | ⑳ Resistivity Sensor | ②⑧ UF Cartridge |
| ⑤ Flow sensor | ⑬ RO Water Outlet | ㉑ Sanitization Block | ②⑨ UP Water Outlet |
| ⑥ PC Pretreatment Cartridge | ⑭ Low tension switch | ㉒ Final Filter | ③⑩ Drain Outlet |
| ⑦ Conductivity Sensor | ⑮ EDI Component | ㉓ DI Water Outlet | |
| ⑧ Pump | ⑯ PE water tank | ㉔ Dispenser arm | |

FXU Specifications

Name	Standard	Low TOC	Eliminating endotoxin	Synthesizing
Model	FXU-60/120/180/250	FXU-60/120/180/250UV	FXU-60/120/180/250UF	FXU-60/120/180/250UVF
Production rate ^[1]	60 series: 60 L/hour, 120 series: 120 L/hour, 180 series: 180 L/hour, 250 series: 250 L/hour			
Dispensing rate ^[2]	Up to 4.2 liters/minute	Up to 4.2 liters/minute	Up to 4.2 liters/minute	Up to 4.2 liters/minute
Ultrapure water quality ^[3]				
Resistivity (25°C) ^[4]	18.2 MΩ.cm	18.2 MΩ.cm	18.2 MΩ.cm	18.2 MΩ.cm
Conductivity (25°C)	0.055 μs/cm	0.055 μs/cm	0.055 μs/cm	0.055 μs/cm
TOC ^[5]	10 ppb ^[6]	3 ppb ^[7]	10 ppb ^[6]	3 ppb ^[7]
Particles ^[8]	<1 /ml (>0.2μm)	<1 /ml (>0.2μm)	<1 /ml (>0.2μm)	<1 /ml (>0.2μm)
Bacteria ^[9]	<0.01 CFU/ml	<0.01 CFU/ml	<0.01 CFU/ml	<0.01 CFU/ml
Endotoxin ^[10]	N/A	N/A	<0.001 EU/ml	<0.001 EU/ml
RNases ^[10]	N/A	N/A	1 pg/ml	1 pg/ml
DNases ^[10]	N/A	N/A	5 pg/ml	5 pg/ml
Protease ^[10]	N/A	N/A	0.15 μg/ml	0.15 μg/ml
DI water quality ^[3]				
Resistivity (25°C) ^[4]	>17.5 MΩ.cm	>17.5 MΩ.cm	>17.5 MΩ.cm	>17.5 MΩ.cm
Conductivity (25°C)	<0.057 μs/cm	<0.057 μs/cm	<0.057 μs/cm	<0.057 μs/cm
Particles ^[8]	<1 /ml (>0.2μm)	<1 /ml (>0.2μm)	<1 /ml (>0.2μm)	<1 /ml (>0.2μm)
Bacteria ^[9]	<0.01 CFU/ml	<0.01 CFU/ml	<0.01 CFU/ml	<0.01 CFU/ml
RO ^{1st} water quality ^[3]				
Ion rejection rate	98%-99% (with new RO module)	98%-99% (with new RO module)	98%-99% (with new RO module)	98%-99% (with new RO module)
Organic rejection rate	>99% (MW>300 Dalton)	>99% (MW>300 Dalton)	>99% (MW>300 Dalton)	>99% (MW>300 Dalton)
Particles and bacteria rejection rate	>99%	>99%	>99%	>99%
Feed water requirements				
Water source type	Tap water	Tap water	Tap water	Tap water
Pressure	1-6 bar	1-6 bar	1-6 bar	1-6 bar
Temperature	5-40 °C	5-40 °C	5-40 °C	5-40 °C
Conductivity	<2000 μs/cm	<2000 μs/cm	<2000 μs/cm	<2000 μs/cm
Total hardness (In CaCO ₃)	<300 ppm	<300 ppm	<300 ppm	<300 ppm
TOC	<2000 ppb	<2000 ppb	<2000 ppb	<2000 ppb
Free chlorine	<3 ppm	<3 ppm	<3 ppm	<3 ppm
PH	4-10	4-10	4-10	4-10
Dissolved CO ₂	<40 ppm	<40 ppm	<40 ppm	<40 ppm
Power supply	220V, 50/60Hz	220V, 50/60Hz	220V, 50/60Hz	220V, 50/60Hz
Total Power	60series: 240W; 120series: 300W; 180series: 500W; 250series: 670W			
Dimension (L×W×H)	607×722×1705mm	607×722×1705mm	607×722×1705mm	607×722×1705mm
Weight	About 118kg	About 118kg	About 118kg	About 118kg
Standard configuration	Main host 1 set All cartridges 1 set 200-liter PE water tank 1 set External leak alarm 1 set	Main host 1 set All cartridges 1 set 200-liter PE water tank 1 set External leak alarm 1 set	Main host 1 set All cartridges 1 set 200-liter PE water tank 1 set External leak alarm 1 set	Main host 1 set All cartridges 1 set 200-liter PE water tank 1 set External leak alarm 1 set

[1] Affected by inlet water quality, pressure, temperature and status of RO membrane

[2] Affected by the tank status and terminal filter

[3] The following values are typical and may vary depending on the nature and concentration of feed water contaminants

[4] According to USP requirements, the resistivity can be displayed as a non-temperature-compensated value

[5] Affected by the type of organics

[6] Inlet TOC<1000ppb, follow professional operating procedures and correct sampling conditions

[7] Inlet TOC<50ppb, follow professional operating procedures and correct sampling conditions

[8] Equip with terminal microfilter and follow professional operating procedures and correct sampling conditions

[9] Equip with terminal microfilter and follow professional operating procedures and correct sampling conditions

[10] Equip with terminal ultrafilter and follow professional operating procedures and correct sampling conditions

FXUS Flagship series

Intelligent Medium Ultrapure Water System

—Ultrapure water, high pure water, and RO^{2nd} water

Integrated all-plastic box body, with a built-in professional-grade 200-liter PE pure water tank.

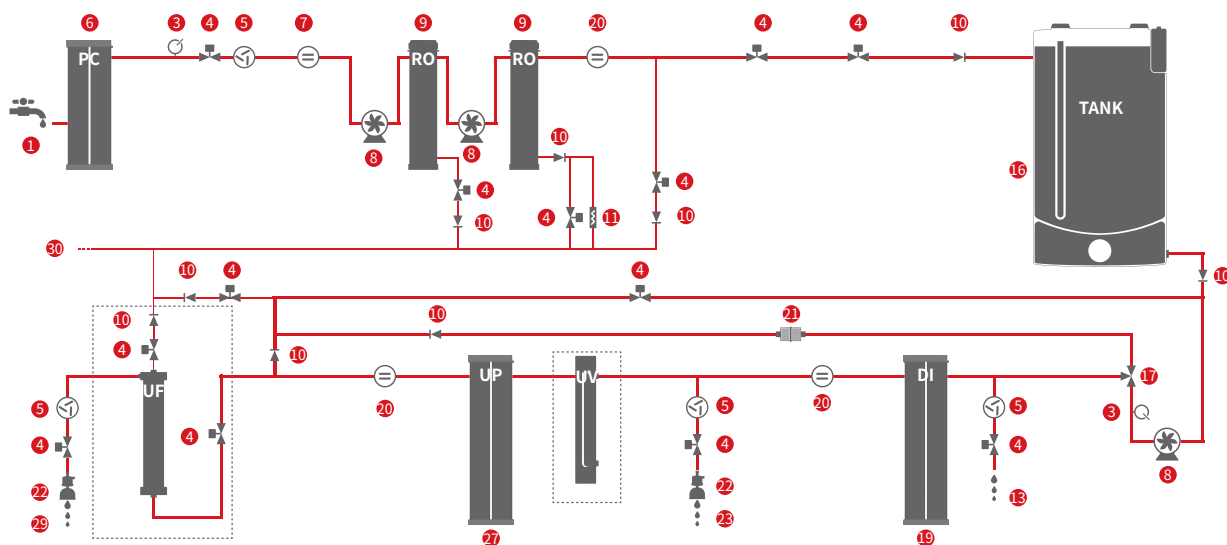
With tap water inlet, using the innovative intelligent human-computer interactive control system and 7-inch colorful capacitive touch screen, integrating functions of Internet of Things (IOT) and cloud platform, embedding new purification cartridges with patented structure, rigorous double RO system, and DI ion-exchange cartridges with larger capacity.

System output: 50, 100, 150 liters/h. It can simultaneously produce ultrapure water (18.2MΩ.cm), high pure water (>17.5MΩ.cm) and double RO water (<5μs/cm). The quality of pure water fully meets or exceeds the requirements of water quality standard specified by ASTM D1193-06, GB/T 11446.1-2013, GB/T 33087-2016, GB/T 6682-2008, CP, EP, USP, JP, CAP, CLSI, etc.

Optional PWS-intelligent storage and distribution system module, compatible with various accessories, delivers pure water through lab pure water pipelines at specific flow rates and pressures, meeting the overall pure water needs of multiple lab or entire lab building.



Flow Diagram



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|-----------------------------|-----------------------|-----------------------|--------------------------|
| ① Feed Water | ⑨ RO cartridge | ⑰ Three way valve | ⑳ UV Component |
| ② PP Pretreatment Cartridge | ⑩ One way valve | ⑱ High tension switch | ㉑ TOC Component |
| ③ Pressure sensor | ⑪ Flow Restrictor | ⑲ DI Cartridge | ㉒ UP Ultrapure cartridge |
| ④ Solenoid valve | ⑫ Pressure water tank | ㉓ Resistivity Sensor | ㉔ UF Cartridge |
| ⑤ Flow sensor | ⑬ RO Water Outlet | ㉔ Sanitization Block | ㉕ UP Water Outlet |
| ⑥ PC Pretreatment Cartridge | ⑭ Low tension switch | ㉕ Final Filter | ㉖ Drain Outlet |
| ⑦ Conductivity Sensor | ⑮ EDI Component | ㉖ DI Water Outlet | |
| ⑧ Pump | ⑯ PE water tank | ㉗ Dispenser arm | |

FXUS Specifications

Name	Standard	Low TOC	Eliminating endotoxin	Synthesizing
Model	FXUS-50/100/150	FXUS-50/100/150UV	FXUS-50/100/150UF	FXUS-50/100/150UVF
Production rate ^[1]	50series: 50 L/hour; 100series: 100 L/hour; 150series: 150 L/hour			
Dispensing rate ^[2]	Up to 2.5 liters/minute	Up to 2.5 liters/minute	Up to 2.5 liters/minute	Up to 2.5 liters/minute
Ultrapure water quality ^[3]				
Resistivity (25°C) ^[4]	18.2 MΩ.cm	18.2 MΩ.cm	18.2 MΩ.cm	18.2 MΩ.cm
Conductivity (25°C)	0.055 μs/cm	0.055 μs/cm	0.055 μs/cm	0.055 μs/cm
TOC ^[5]	10 ppb ^[6]	3 ppb ^[7]	10 ppb ^[6]	3 ppb ^[7]
Particles ^[8]	<1 /ml (>0.2μm)	<1 /ml (>0.2μm)	<1 /ml (>0.2μm)	<1 /ml (>0.2μm)
Bacteria ^[9]	<0.01 CFU/ml	<0.01 CFU/ml	<0.01 CFU/ml	<0.01 CFU/ml
Endotoxin ^[10]	N/A	N/A	<0.001 EU/ml	<0.001 EU/ml
RNases ^[10]	N/A	N/A	1 pg/ml	1 pg/ml
DNases ^[10]	N/A	N/A	5 pg/ml	5 pg/ml
Protease ^[10]	N/A	N/A	0.15 μg/ml	0.15 μg/ml
DI water quality ^[3]				
Resistivity (25°C) ^[4]	>17.5 MΩ.cm	>17.5 MΩ.cm	>17.5 MΩ.cm	>17.5 MΩ.cm
Conductivity (25°C)	<0.057 μs/cm	<0.057 μs/cm	<0.057 μs/cm	<0.057 μs/cm
Particles ^[8]	<1 /ml (>0.2μm)	<1 /ml (>0.2μm)	<1 /ml (>0.2μm)	<1 /ml (>0.2μm)
Bacteria ^[9]	<0.01 CFU/ml	<0.01 CFU/ml	<0.01 CFU/ml	<0.01 CFU/ml
RO ^{2nd} water quality ^[3]				
Resistivity (25°C) ^[4]	>0.2 MΩ.cm	>0.2 MΩ.cm	>0.2 MΩ.cm	>0.2 MΩ.cm
Conductivity (25°C)	<5 μs/cm	<5 μs/cm	<5 μs/cm	<5 μs/cm
Organic rejection rate	>99% (MW>300 Dalton)	>99% (MW>300 Dalton)	>99% (MW>300 Dalton)	>99% (MW>300 Dalton)
Particles and bacteria rejection rate	>99%	>99%	>99%	>99%
Feed water requirements				
Water source type	Tap water	Tap water	Tap water	Tap water
Pressure	1-6 bar	1-6 bar	1-6 bar	1-6 bar
Temperature	5-40 °C	5-40 °C	5-40 °C	5-40 °C
Conductivity	<2000 μs/cm	<2000 μs/cm	<2000 μs/cm	<2000 μs/cm
Total hardness (In CaCO ₃)	<300 ppm	<300 ppm	<300 ppm	<300 ppm
TOC	<2000 ppb	<2000 ppb	<2000 ppb	<2000 ppb
Free chlorine	<3 ppm	<3 ppm	<3 ppm	<3 ppm
PH	4-10	4-10	4-10	4-10
Dissolved CO ₂	<40 ppm	<40 ppm	<40 ppm	<40 ppm
Power supply	220V, 50/60Hz	220V, 50/60Hz	220V, 50/60Hz	220V, 50/60Hz
Total Power	50series: 240W; 100series: 480W; 150series: 850W			
Dimension (L×W×H)	607×722×1705mm	607×722×1705mm	607×722×1705mm	607×722×1705mm
Weight	About 137kg	About 137kg	About 137kg	About 137kg
Standard configuration	Main host 1 set All cartridges 1 set 200-liter PE water tank 1 set External leak alarm 1 set	Main host 1 set All cartridges 1 set 200-liter PE water tank 1 set External leak alarm 1 set	Main host 1 set All cartridges 1 set 200-liter PE water tank 1 set External leak alarm 1 set	Main host 1 set All cartridges 1 set 200-liter PE water tank 1 set External leak alarm 1 set

[1] Affected by inlet water quality, pressure, temperature and status of RO membrane

[2] Affected by the tank status and terminal filter

[3] The following values are typical and may vary depending on the nature and concentration of feed water contaminants

[4] According to USP requirements, the resistivity can be displayed as a non-temperature-compensated value

[5] Affected by the type of organics

[6] Inlet TOC<1000ppb, follow professional operating procedures and correct sampling conditions

[7] Inlet TOC<50ppb, follow professional operating procedures and correct sampling conditions

[8] Equip with terminal microfilter and follow professional operating procedures and correct sampling conditions

[9] Equip with terminal microfilter and follow professional operating procedures and correct sampling conditions

[10] Equip with terminal ultrafilter and follow professional operating procedures and correct sampling conditions

FXUE Flagship series

Intelligent Medium Ultrapure Water System

—Ultrapure water, EDI water, and RO^{2nd} water

Integrated all-plastic box body, with a built-in professional-grade 200-liter PE pure water tank.

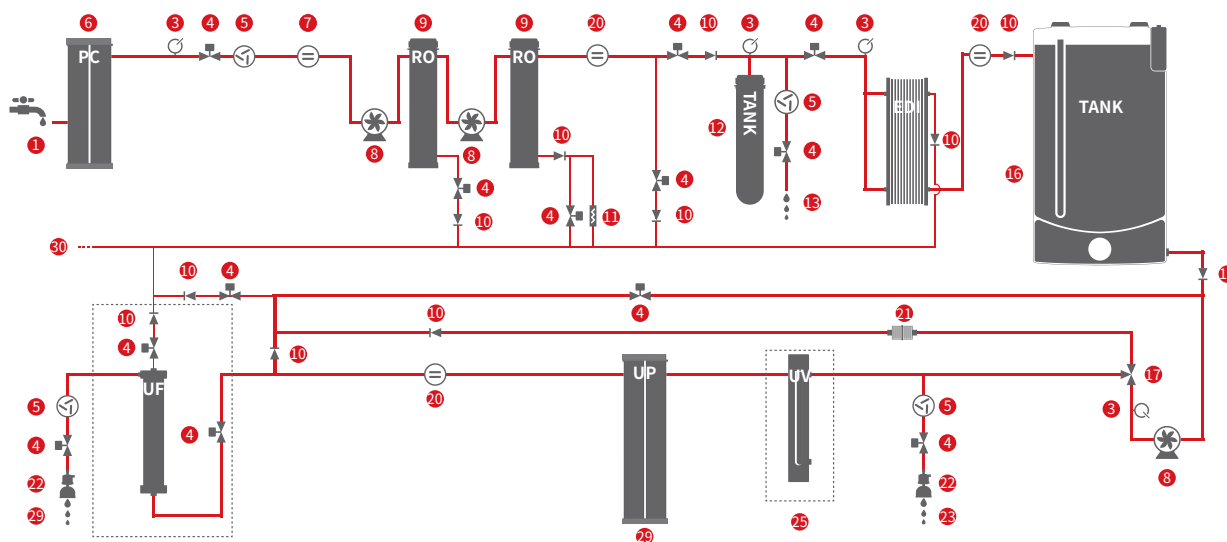
With tap water inlet, using the innovative intelligent human-computer interactive control system and 7-inch colorful capacitive touch screen, integrating functions of Internet of Things (IOT) and cloud platform, embedding new purification cartridges with patented structure, rigorous double RO system, advanced EDI module and DI ion-exchange cartridges with larger capacity.

System output: 50, 100, 150 liters/h. Maximum output per day is up to 3600 liters. It can simultaneously produce ultrapure water (18.2MΩ.cm), EDI water (Resistivity>15MΩ.cm, TOC<30ppb) and double RO water(<5μs/cm) with optimized running cost. The quality of pure water fully meets or exceeds the requirements of water quality standard specified by ASTM D1193-06, GB/T 11446.1-2013, GB/T 33087-2016, GB/T 6682-2008, CP, EP, USP, JP, CAP, CLSI, etc.

Optional PWS-intelligent storage and distribution system module, compatible with various accessories, delivers pure water through lab pure water pipelines at specific flow rates and pressures, meeting the overall pure water needs of multiple lab or entire lab building.



Flow Diagram



- | | | | |
|-----------------------------|-----------------------|-----------------------|---------------------------|
| ① Feed Water | ⑨ RO cartridge | ⑰ Three way valve | ②⑤ UV Component |
| ② PP Pretreatment Cartridge | ⑩ One way valve | ⑱ High tension switch | ②⑥ TOC Component |
| ③ Pressure sensor | ⑪ Flow Restrictor | ⑲ DI Cartridge | ②⑦ UP Ultrapure cartridge |
| ④ Solenoid valve | ⑫ Pressure water tank | ⑳ Resistivity Sensor | ②⑧ UF Cartridge |
| ⑤ Flow sensor | ⑬ RO Water Outlet | ㉑ Sanitization Block | ②⑨ UP Water Outlet |
| ⑥ PC Pretreatment Cartridge | ⑭ Low tension switch | ㉒ Final Filter | ③⑩ Drain Outlet |
| ⑦ Conductivity Sensor | ⑮ EDI Component | ㉓ DI Water Outlet | |
| ⑧ Pump | ⑯ PE water tank | ㉔ Dispenser arm | |

FXUE Specifications

Name	Standard	Low TOC	Eliminating endotoxin	Synthesizing
Model	FXUE-50/100/150	FXUE-50/100/150UV	FXUE-50/100/150UF	FXUE-50/100/150UVF
Production rate ^[1]	50series: 50 L/hour; 100series: 100 L/hour; 150series: 150 L/hour			
Dispensing rate ^[2]	Up to 2.5 liters/minute	Up to 2.5 liters/minute	Up to 2.5 liters/minute	Up to 2.5 liters/minute
Ultrapure water quality ^[3]				
Resistivity (25°C) ^[4]	18.2 MΩ.cm	18.2 MΩ.cm	18.2 MΩ.cm	18.2 MΩ.cm
Conductivity (25°C)	0.055 μs/cm	0.055 μs/cm	0.055 μs/cm	0.055 μs/cm
TOC ^[5]	10 ppb ^[6]	3 ppb ^[7]	10 ppb ^[6]	3 ppb ^[7]
Particles ^[8]	<1 /ml (>0.2μm)	<1 /ml (>0.2μm)	<1 /ml (>0.2μm)	<1 /ml (>0.2μm)
Bacteria ^[9]	<0.01 CFU/ml	<0.01 CFU/ml	<0.01 CFU/ml	<0.01 CFU/ml
Endotoxin ^[10]	N/A	N/A	<0.001 EU/ml	<0.001 EU/ml
RNases ^[10]	N/A	N/A	1 pg/ml	1 pg/ml
DNases ^[10]	N/A	N/A	5 pg/ml	5 pg/ml
Protease ^[10]	N/A	N/A	0.15 μg/ml	0.15 μg/ml
EDI water quality ^[3]				
Resistivity (25°C) ^[4]	>15 MΩ.cm	>15 MΩ.cm	>15 MΩ.cm	>15 MΩ.cm
Conductivity (25°C)	<0.067 μs/cm	<0.067 μs/cm	<0.067 μs/cm	<0.067 μs/cm
TOC ^[5]	≤ 30 ppb	≤ 30 ppb	≤ 30 ppb	≤ 30 ppb
Particles ^[8]	<1 /ml (>0.2μm)	<1 /ml (>0.2μm)	<1 /ml (>0.2μm)	<1 /ml (>0.2μm)
Bacteria ^[9]	<0.01 CFU/ml	<0.01 CFU/ml	<0.01 CFU/ml	<0.01 CFU/ml
RO ^{2nd} water quality ^[3]				
Resistivity (25°C) ^[4]	>0.2 MΩ.cm	>0.2 MΩ.cm	>0.2 MΩ.cm	>0.2 MΩ.cm
Conductivity (25°C)	<5 μs/cm	<5 μs/cm	<5 μs/cm	<5 μs/cm
Organic rejection rate	>99% (MW>300 Dalton)	>99% (MW>300 Dalton)	>99% (MW>300 Dalton)	>99% (MW>300 Dalton)
Particles and bacteria rejection rate	>99%	>99%	>99%	>99%
Feed water requirements				
Water source type	Tap water	Tap water	Tap water	Tap water
Pressure	1-6 bar	1-6 bar	1-6 bar	1-6 bar
Temperature	5-40 °C	5-40 °C	5-40 °C	5-40 °C
Conductivity	<2000 μs/cm	<2000 μs/cm	<2000 μs/cm	<2000 μs/cm
Total hardness (In CaCO ₃)	<300 ppm	<300 ppm	<300 ppm	<300 ppm
TOC	<2000 ppb	<2000 ppb	<2000 ppb	<2000 ppb
Free chlorine	<3 ppm	<3 ppm	<3 ppm	<3 ppm
PH	4-10	4-10	4-10	4-10
Dissolved CO ₂	<40 ppm	<40 ppm	<40 ppm	<40 ppm
Power supply	220V, 50/60Hz	220V, 50/60Hz	220V, 50/60Hz	220V, 50/60Hz
Total Power	50series: 276W; 100series: 536W; 150series: 886W			
Dimension (L×W×H)	607×722×1705mm	607×722×1705mm	607×722×1705mm	607×722×1705mm
Weight	About 140kg	About 140kg	About 140kg	About 140kg
Standard configuration	Main host 1 set All cartridges 1 set 200-liter PE water tank 1 set 1.8-liter pressure water tank 1 set External leak alarm 1 set	Main host 1 set All cartridges 1 set 200-liter PE water tank 1 set 1.8-liter pressure water tank 1 set External leak alarm 1 set	Main host 1 set All cartridges 1 set 200-liter PE water tank 1 set 1.8-liter pressure water tank 1 set External leak alarm 1 set	Main host 1 set All cartridges 1 set 200-liter PE water tank 1 set 1.8-liter pressure water tank 1 set External leak alarm 1 set

[1] Affected by inlet water quality, pressure, temperature and status of RO membrane

[2] Affected by the tank status and terminal filter

[3] The following values are typical and may vary depending on the nature and concentration of feed water contaminants

[4] According to USP requirements, the resistivity can be displayed as a non-temperature-compensated value

[5] Affected by the type of organics

[6] Inlet TOC<1000ppb, follow professional operating procedures and correct sampling conditions

[7] Inlet TOC<50ppb, follow professional operating procedures and correct sampling conditions

[8] Equip with terminal microfilter and follow professional operating procedures and correct sampling conditions

[9] Equip with terminal microfilter and follow professional operating procedures and correct sampling conditions

[10] Equip with terminal ultrafilter and follow professional operating procedures and correct sampling conditions

FXDE Flagship series

Intelligent Medium Pure Water System

—EDI water, RO^{2nd} water

Integrated all-plastic box body, with a built-in professional-grade 200-liter PE pure water tank.

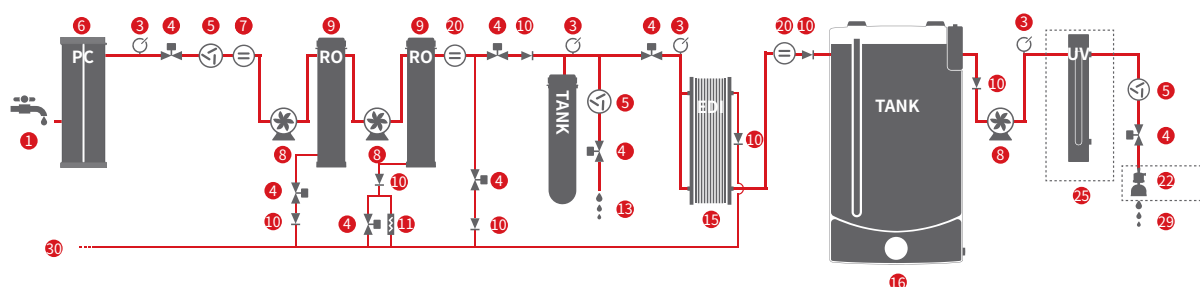
With tap water inlet, using the innovative intelligent human-computer interactive control system and 7-inch colorful capacitive touch screen, integrating functions of Internet of Things (IOT) and cloud platform, embedding new purification cartridges with patented structure, rigorous double RO system, and advanced EDI module.

System output: 50, 100, 150 liters/h. Maximum output per day is up to 3600 liters. It can simultaneously produce EDI water (Resistivity>15MΩ.cm, TOC<30ppb) and double RO water(<5μs/cm) with optimized running cost. The quality of pure water fully meets or exceeds the requirements of water quality standard specified by ISO3696 (Grade 2), GB/T 6682 (Grade 1), ASTM D1193 (Type II reagent water), JIS K0557, etc., also meets the purified water technical requirements of CP, EP, USP, JP and other national pharmacopoeia.

Optional PWS-intelligent storage and distribution system module, compatible with various accessories, delivers pure water through lab pure water pipelines at specific flow rates and pressures, meeting the overall pure water needs of multiple lab or entire lab building.



Flow Diagram



- | | | | |
|-----------------------------|-----------------------|-----------------------|--------------------------|
| ① Feed Water | ⑨ RO cartridge | ⑰ Three way valve | ⑳ UV Component |
| ② PP Pretreatment Cartridge | ⑩ One way valve | ⑱ High tension switch | ㉑ TOC Component |
| ③ Pressure sensor | ⑪ Flow Restrictor | ㉒ DI Cartridge | ㉓ UP Ultrapure cartridge |
| ④ Solenoid valve | ⑫ Pressure water tank | ㉔ Resistivity Sensor | ㉕ UF Cartridge |
| ⑤ Flow sensor | ⑬ RO Water Outlet | ㉖ Sanitization Block | ㉗ UP Water Outlet |
| ⑥ PC Pretreatment Cartridge | ⑭ Low tension switch | ㉘ Final Filter | ㉙ Drain Outlet |
| ⑦ Conductivity Sensor | ⑮ EDI Component | ㉚ DI Water Outlet | |
| ⑧ Pump | ⑯ PE water tank | ㉛ Dispenser arm | |

FXDE Specifications

Name	Standard	Eliminating bacteria and particle
Model	FXDE-50/100/150	FXDE-50/100/150UT
Production rate ^[1]	50series: 50 L/hour; 100series: 100 L/hour; 150series: 150 L/hour	
Dispensing rate ^[2]	Up to 2.5 liters/minute	Up to 2.5 liters/minute
EDI water quality ^[3]		
Resistivity (25°C) ^[4]	>15 MΩ.cm	>15 MΩ.cm
Conductivity (25°C)	<0.067 μs/cm	<0.067 μs/cm
TOC ^[5]	≤ 30 ppb	≤ 30 ppb
Particles ^[8]	N/A	<1 /ml (>0.2μm)
Bacteria ^[9]	N/A	<0.01 CFU/ml
RO ^{2nd} water quality ^[3]		
Resistivity (25°C) ^[4]	>0.2 MΩ.cm	>0.2 MΩ.cm
Conductivity (25°C)	<5 μs/cm	<5 μs/cm
Organic rejection rate	>99% (MW>300 Dalton)	>99% (MW>300 Dalton)
Particles and bacteria rejection rate	>99%	>99%
Feed water requirements		
Water source type	Tap water	Tap water
Pressure	1-6 bar	1-6 bar
Temperature	5-40 °C	5-40 °C
Conductivity	<2000 μs/cm	<2000 μs/cm
Total hardness (In CaCO ₃)	<300 ppm	<300 ppm
TOC	<2000 ppb	<2000 ppb
Free chlorine	<3 ppm	<3 ppm
PH	4-10	4-10
Dissolved CO ₂	<40 ppm	<40 ppm
Power supply	220V, 50/60Hz	220V, 50/60Hz
Total Power	50series: 276W; 100series: 536W; 150series: 866W	50series: 276W; 100series: 536W; 150series: 866W
Dimension (L×W×H)	607×722×1705mm	607×722×1705mm
Weight	About 131kg	About 131kg
Standard configuration	Main host 1 set All cartridges 1 set 200-liter PE water tank 1 set 1.8-liter pressure water tank 1 set External leak alarm 1 set	Main host 1 set All cartridges 1 set 200-liter PE water tank 1 set 1.8-liter pressure water tank 1 set External leak alarm 1 set

[1] Affected by inlet water quality, pressure, temperature and status of RO membrane

[2] Affected by the tank status and terminal filter

[3] The following values are typical and may vary depending on the nature and concentration of feed water contaminants

[4] According to USP requirements, the resistivity can be displayed as a non-temperature-compensated value

[5] Affected by the type of organics

[6] Inlet TOC<1000ppb, follow professional operating procedures and correct sampling conditions

[7] Inlet TOC<50ppb, follow professional operating procedures and correct sampling conditions

[8] Equip with terminal microfilter and follow professional operating procedures and correct sampling conditions

[9] Equip with terminal microfilter and follow professional operating procedures and correct sampling conditions

[10] Equip with terminal ultrafilter and follow professional operating procedures and correct sampling conditions

Hyperpurex® lab water system

Bring you products and services beyond expected

ISO
3696
US Pharmacopoeia
GB/T 33087 2016
Japan Pharmacopoeia
ISO9001
CLSI GB/T.11446 1-2013
ASTM D 5196 **ISO14001**
China Pharmacopoeia ASTM
GB/T 6682-2008 JIS K 0557
Eu Pharmacopoeia D1193
CE Quality Standard
HyperpureX®

PRODUCT

- Under management system of ISO9001 and ISO14001, in accordance with CE quality standards, we carry out product design, research & development and manufacturing to ensure long-term stability and reliability of quality.
- To help you meet industry specifications, we can assist in providing certificates of conformity, calibration certificates, quality certificates, performance reports, water quality compliance certificates and other supporting documents upon request.
- Hyperpurex® FX Flagship series - lab water system can produce pure water/ultrapure water to meet the requirements of the following organizations:
- Chinese Pharmacopoeia-CP, United States Pharmacopoeia-USP, European Pharmacopoeia-EP, Japanese Pharmacopoeia -JP, GB/T 33087-2016, GB/T 6682-2008, GB/T 11446.1-2013, ASTM D1193, ASTM D 5196, ISO 3696, CLSI, JIS K 0557.

SERVICE

We wholeheartedly serve, only for your full satisfaction.

With customer satisfaction as the service goal, to continue to create value for customers as the direction, to grow together with customers as the concept, based on professionalism, we are full of sincerity and enthusiasm, committing to providing customers with professional and perfect technical support and after-sales service. So that you can devote all your energy to focus on the work.

Our service include:

- 24 months product warranty (excluding filter consumables)
- On-site professional training of installation, use and maintenance.
- Regular engineer return visit service
- Free continuous optimization and upgrading service of product life cycle.
- 24/7 technical support hotline (4000701800) and Wechat service number.
- Professional and rigorous 3Q(IQ/OQ/PQ) verification documentation and verification services in both English and Chinese, to help you meet compliance requirements of GLP, GMP and cGMP.

Ordering Information

[illegible]

Ordering Information

Cartridge	Item No	Product description	Item No	Product description
	HPC104	Pretreatment cartridge D	HPC603	UP cartridge, standard
	HPC360	RO ^{1st} module S60	HPC604	UP cartridge, Low TOC
	HPC312	RO ^{1st} module S120	HPC605	UP cartridge, standard
	HPC318	RO ^{1st} module S180	HPC606	UP cartridge, Low TOC
	HPC325	RO ^{1st} module S250	HPC710	Air filter for tank
	HPC350	RO ^{1st} module F50	HPC703	185&254nm double wavelength UV lamp
	HPC310	RO ^{1st} module F100	HPC704	254nm UV lamp
	HPC315	RO ^{1st} module F150	HPC709	UF ultrafiltration module
	HPC450	RO ^{2nd} module D50	HPC801	TF terminal microfilter
	HPC410	RO ^{2nd} module D100	HPC802	TF terminal microfilter
	HPC415	RO ^{2nd} module D150	HPC803	TF terminal microfilter
	HPC502	DI cartridge	HPC804	TF terminal microfilter
	HPC503	DI cartridge	HPC810	UF terminal ultrafilter
Accessory	Item No	Product description	Item No	Product description
	DISP2000	HiDis dispenser arm (independent), equipped with 2M connection kit	PWA7502	External leak sensor
	PWA7201	Automatic water softener (salt required)	PWSL1000	Pure water supplying module (BA)
	PWA7020	Pretreatment filter for source water	PWSL2000	Pure water storage and supplying module (PR)
	PWA7021	PP cartridge for pretreatment filter	PWSM0500	Pure water storage and supplying module (FC-500L)
	PWA7022	RS cartridge for pretreatment filter	PWSM1000	Pure water storage and supplying module (FC-1000L)
	PWA7501	Foot switch	PWSM2000	Pure water storage and supplying module (FC-2000L)
Service	Item No	Product description		
	HPS51001	1-year extended warranty service (except for consumables)		
	HPS51003	3-year extended warranty service (except for consumables)		
	HPS52001	Verification documents in English		
	HPS53001	Basic verification service		
	HPS59001*	1-year, one-price all-inclusive maintenance agreement, including regular consumables replacement, maintenance and calibration		
	HPS59003*	3-year, one-price all-inclusive maintenance agreement, including regular consumables replacement, maintenance and calibration		

*On the basis of mutual confirmation of pure water consumption and feed water quality.



Web YouTube Instagram

For more product details, please login: www.hpurex.com

Tel: 0086-21-3107 5991

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